Cumulative Examination
Analytical Chemistry
Friday, April 27, 2012

1. Explain the difference between a fluorescence emission spectrum and a fluorescence excitation spectrum. Which more closely resembles an absorption spectrum and why? **(20 points)**

2. Define the following terms: (a) fluorescence, (b) phosphorescence, (c) resonance fluorescence, (d) singlet state, (e) triplet state, (f) vibrational relaxation, (g) internal conversion, (h) external conversion, (i) intersystem crossing, (j) predissociation, (k) dissociation, (l) quantum yield, (m) chemiluminescence **(26 points)**

3. Why is spectrofluorometry potentially more sensitive than spectrophotometry? **(20 points)**

4. Why do some absorbing compounds fluoresce, but others do not? **(20 points)**

5. Quinine in a 1.664 g antimalarial tablet was dissolved in sufficient 0.10 M HCl to give 500 mL of solution. A 20.0 mL aliquot was then diluted to 100.0 mL with the acid. The fluorescence intensity for the diluted sample at 347.5 nm provided a reading of 245 on an arbitrary scale. A standard 100 ppm quinine solution registered 125 when measured under conditions identical to those for the diluted sample. Calculate the mass in milligrams of quinine in the tablet. **(14 points)**